STA380

TUT12 Version B

Question 2

(a) **[3 points]** Assuming a random sample from the Gaussian distribution with mean μ and standard deviation σ, derive the log likelihood function, $l(\mu, \sigma, \mathbf{x})$.

(b) **[3 points]** Assume that $\mu = 2$ and realisations $X_1 = x_1, \ldots, X_n = x_n$ are known and stored in a vector called `x`. Use `optimize()` to maximise the likelihood function with respect to σ. Explicitly write out the `R` code.

(c) **[4 points]** Assume that μ and σ are unknown, but the realisations $X_1 = x_1, \ldots, X_n = x_n$ are known and stored in a vector called `x`. Use `optim()` and the Nelder-Mead method to maximise the likelihood function to determine the value of μ and σ. Explicitly write out the `R` code.

(a) pdf : $f(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

① Likelihood.

$$L(\mu, \sigma \mid x) = \underbrace{\left(\frac{1}{\sqrt{2\pi} \cdot \sigma}\right)^n}_{} * \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right)$$

② log_likelihood

$$l(\mu, \sigma \mid x) = \underbrace{-\frac{n}{2}\log(2\pi)}_{①} - n\log(\sigma) \underbrace{- \frac{1}{2\sigma^2} \boxed{\sum_{i=1}^{n} (x_i - \mu)^2}}_{②}$$

(b) R codes

```
log_likelihood <- function (sigma, mu, data) {

  n <- length (data)

  part_1 <- - n/2 * log (2*pi) - n * log (sigma)

  part_2 <- -1/(2* sigma^2) * sum ( (data - mu)^2 )

  return ( part_1 + part_2 )

}

optimize ( log_likelihood , lower = 0.00001 , upper = 10, mu=2,
    data= x , maximum = TRUE ) $ maximum
```

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default

```
neg_log_likelihood <- function ( theta, data ) {
  mu <- theta [1]
  sigma <- theta [2]
  if (sigma <= 0) return (-Inf)
  -1 * log_likelihood ( sigma, mu, data )
}.

optim (c(2, 3), neg_log_likelihood, data = x, ) $ par
```

method = "Nelder-Mead"

Question 3

(a) **[4 points]** Let $X_1, X_2, \ldots, X_n$ be a random sample with realisations $x_1, x_2, \ldots, x_n$. Consider the following estimate for $\theta := \mathbb{E}[X^2]$:

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i^2$$

Prove that the jackknife estimate for θ is equal to $\hat{\theta}$, where the jackknife estimate is $\hat{\theta}_{(.)}$ as defined below:

$$\hat{\theta}_{(.)} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(i)}, \quad \hat{\theta}_{(i)} = \frac{1}{n-1}\sum_{j=1, j\neq i}^{n} x_j^2$$

$i = 3$

$j = 1, 2, 3 \ldots n$

Proof:

$$\hat{\theta}_{.} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(i)}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\frac{1}{n-1}\sum_{j=1, j\neq i}^{n} x_j^2$$

$$= \frac{1}{n*(n-1)}\sum_{j=1}^{n}(n-1)\, x_j^2$$

$$= \frac{\cancel{n-1}}{n*\cancel{(n-1)}}\sum_{j=1}^{n} x_j^2$$

$$= \frac{1}{n}\sum_{j=1}^{n} x_j^2$$

if $i=2$ $\quad x_1^2 + x_2^2 + x_3^2 - x_2^2$

$$\sum_{i=j}^{n} x_j^2 - x_i^2$$

Toy Example:

if $n=3$

$i=1$, $j=2, 3$ $\quad x_2^2 + x_3^2$

$i=2$, $j=1, 3$ $\quad x_1^2 + x_3^2$

$i=3$, $j=1, 2$ $\quad x_1^2 + x_2^2$

$$2x_1^2 + 2x_2^2 + 2x_3^2$$

$$= \sum_{j=1}^{3} 2\, x_j^2$$

$$= \sum_{j=1}^{n} (n-1)\, x_j^2$$

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i^2$$

Conclusion $\quad \hat{\theta}_{(.)} = \frac{1}{n}\sum_{j=1}^{n} x_j^2 = \hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i^2$

(b) **[6 points]** Let $k \in \mathbb{N}$. Consider the following estimate for $\theta := \mathbb{E}[X^k]$:

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n} x_i^k$$

Is it the case that for any k, the jackknife estimate for θ is equal to $\hat{\theta}$? Prove this or provide a counterexample. Recall that the jackknife estimate is $\hat{\theta}_{(.)}$ as defined below:

$$\hat{\theta}_{(.)} = \frac{1}{n}\sum_{i=1}^{n} \hat{\theta}_{(i)}, \quad \hat{\theta}_{(i)} = \frac{1}{n-1}\sum_{j=1, j\neq i}^{n} x_j^k.$$

Yes

Question 5

Estimate

$$\theta = \int_0^1 \tan(x) e^{-x^2/2} dx$$

using importance sampling, where the importance function is the pdf of the standard Gaussian distribution. Assume that you need $n = 10^4$ replicates.

To generate a random sample from the Normal distribution, one could use the `rnorm(n, mean = 0, sd = 1)` function, where `n` represents the number of observations, `mean` represents the mean, and `sd` represents the standard deviation.

(a) **[5 points]** Write the algorithm using importance sampling. **Show your work.**

(b) **[2 points]** Write out the `R` code for the algorithm using importance sampling. Assume we want $n = 10^4$ replicates.

(a). $f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

Normal (mean=0, sd=1) $\Rightarrow f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$

pdf Normal(0, 1)

$$\theta = \int_0^1 \tan(x) \cdot e^{-\frac{x^2}{2}} * \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}} dx.$$

$$= \int_0^1 \tan(x) \sqrt{2\pi} * f_{N(0,1)}(x)\, dx$$

$$= E\left[\sqrt{2\pi} * \tan(X) * I\{0 \le X \le 1\}\right]$$

Algorithm: (1) Let $x_1, \dots x_{10^4}$ be random variables generated from Normal (0, 1)

(2) $\hat{\theta} = \frac{1}{10^4} \sum_{i=1}^{10^4} \sqrt{2\pi} * \tan(x_i) * 1\{0 \le x_i \le 1\}$

(b) R codes

```
n <- 10^4
x <- rnorm (n, mean=0, sd=1)
theta_hat <- mean ( sqrt (2 * pi) * tan(x) * ( x>=0 & x<=1))
```

Question 6

Let $X_1, X_2, \ldots, X_n$ be a random sample with cdf $F_X(x)$. Recall that the ecdf is $F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I(X_i \le x)$. Recall that $F_n(x)$ is an unbiased estimator of $F_X(x)$.

(a) **[3 points]** Compute $\mathbb{V}ar(F_n(x))$.

(b) **[3 points]** Compute $\lim_{n\to\infty} \mathbb{V}ar(F_n(x))$.

(c) **[4 points]** Using Chebychev's inequality, prove that $F_n(x) \xrightarrow{p} F_X(x)$.

Hint. Chebyshev's inequality is the following: for any $\varepsilon > 0$,

$$\mathbb{P}(|Y - \mathbb{E}[Y]| \ge \varepsilon) \le \frac{\mathrm{Var}(Y)}{\varepsilon^2}.$$

For a sequence of random variables $X_1, X_2, \ldots$, we say the sequence converges in probability to X, ($X_n \xrightarrow{p} X$) if

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n > N, \mathbb{P}\{|X_n - X| > \epsilon\} < \epsilon.$$

(a) $Var(F_n(x))$

$$F_n(x) = \sum_{i=1}^{n} I(x_i \le x) \quad \text{ecdf.}$$

$$I(x_i \le x) = \begin{cases} 1, & \text{if } x_i \le x \\ 0, & \text{if } x_i > x \end{cases}$$

$$P(Y_i = 1) = P(X_i \le x) = F_X(x)$$

$Y_i \sim \text{Bernoulli}(p)$ with $p = F_X(x)$

$p(1-p)$

$$Var(F_n(x)) = Var\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right)$$

since Y_i is iid,

$$= \frac{1}{n^2} Var\left(\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2} \cdot n \cdot Var(Y_i)$$

$$= \frac{p(1-p)}{n}$$

$$= \frac{F_X(x) * (1 - F_X(x))}{n}.$$

(b) **[3 points]** Compute $\lim_{n\to\infty} \mathbb{V}ar(F_n(x))$. $\lim_{n\to\infty} Var(F_n(x)) = \lim_{n\to\infty} \frac{F_X(x)\cdot(1-F_X(x))}{n} = 0$

(c) **[4 points]** Using Chebychev's inequality, prove that $F_n(x) \xrightarrow{p} F_X(x)$.

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General.

$$\lim_{n\to\infty} P(|Y - E[Y]| \geq \varepsilon) = 0$$

$$\lim_{n\to\infty} P(|F_n(x) - E(F_n(x))| \geq \varepsilon) \leq \frac{Var(F_n(x))}{\varepsilon^2}$$

$$\leq \frac{F_X(x)(1-F_X(x))}{\underset{\infty}{n} * \varepsilon^2}$$

$$\leq 0$$

Since the probability is always be non-negative.

$$\lim_{n\to\infty} P(|F_n(x) - E(F_n(x))| \geq \varepsilon) = 0$$

$$\Rightarrow F_n(x) \xrightarrow{P} F_X(x)$$